MIHIKA RAO

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PROFESSIONAL SUMMARY

I like the part of the work before the model: deciding what to optimize, what to measure, and what "better" should mean for real users.

Over 7+ years, I've built personalization and analytics systems that connect behavior → decisions → shipped iteration. At an ed-tech startup, I designed a candidate-generation and ranking approach, evaluated it with offline ranking metrics, and validated improvements through A/B tests—driving a 30% lift in 30-day retention. I also build reliable data pipelines and dashboards that make experimentation repeatable, and I communicate results clearly to technical and non-technical stakeholders.

EXPERIENCE

NOVACARDS (434 ACCELERATOR PROGRAM) **Remote**

Data Science Fellow – Personalization *May 2025 - Aug 2025*

- Built and iterated an AI-driven personalization system to connect learners with the right content at the right time
  - Designed a candidate-generation → ranking workflow (rules + ML) to personalize review scheduling and question difficulty from user history
  - Built offline evaluation to compare ranking strategies using relevance-style metrics (e.g., recall@K / MAP / NDCG as applicable) and cohort performance
  - Ran A/B tests with clear success metrics and guardrails; improvements drove a 30% lift in 30-day retention during the pilot
  - Instrumented key learning telemetry (scores, streaks, session length, error patterns) and built automated pipelines across 5K+ users, cutting manual reporting by $\geq$ 50%
  - Created decision dashboards for product/content teams (retention curves, cohorts, most-missed items) to prioritize iteration
- *Recommender Systems / Ranking & Evaluation / A/B Testing / Product Analytics / Cross-functional Collaboration*

UNIVERSITY OF VIRGINIA & NRAO **Charlottesville, VA**

Graduate Research Assistant – Machine Learning & Data Systems *Aug 2021 - Present*

- Led multi-year, ML-driven analytics projects on large observational datasets, building repeatable pipelines and evaluation frameworks to support reliable decision-making
  - Built end-to-end Python data systems to ingest, clean, transform, and analyze multi-GB datasets; automation reduced processing time by 40% vs. prior workflows
  - Developed predictive/statistical models (regression and probabilistic methods) to quantify relationships between key variables under noise and uncertainty
  - Used simulation-based validation (Monte Carlo / Bayesian-style approaches) to stress-test conclusions and compare "what-if" scenarios
  - Created stakeholder-ready summaries and visualizations that turned complex modeling outputs into clear next steps
- Coordinated across a distributed team to deliver published results on shared timelines
  - Collaborated with 20 researchers across 5 institutions; communicated methods and findings to mixed audiences at 6 international and 10 national conferences
- *Modeling Under Uncertainty / Data Pipelines / Validation & Evaluation Mindset / Stakeholder Communication*

PROJECTS

UVA KARSH INSTITUTE OF DEMOCRACY **Charlottesville, VA**
Graduate Student Advocacy Lead – Quantitative Strategy *Jan 2023 - Oct 2023*

- Built a data-driven case for policy change by translating survey + benchmarking data into decision-ready metrics and narratives
  - Analyzed stipend benchmarks across 25 peer institutions and survey responses from 8K+ students to quantify gaps and priority drivers
  - Built clear visualizations linking compensation to cost of living, reported stress, and self-reported productivity
  - Presented findings to university leadership; work contributed to a $10K annual stipend increase (from $26K $\rightarrow$ $36K) impacting 12K+ graduate students
- *Quantitative Storytelling / Benchmarking / Survey Analytics / Stakeholder Influence*

AGNES SCOTT COLLEGE **Atlanta, GA**
Capstone – Econometrics & Predictive Modeling *Jan 2020 - May 2021*

- Led a capstone econometrics project using national labor data to identify key drivers of workplace well-being
  - Cleaned and modeled 50K+ records of U.S. labor microdata (100+ variables) to identify drivers of workplace well-being
  - Built and compared logistic/linear and regularized models; quantified marginal effects of key drivers (e.g., schedule control, benefits coverage) and communicated practical implications
- *Predictive Modeling / Feature Engineering / Model Comparison / Insight Communication*

SKILLS

Programming	Python (NumPy, Pandas, SciPy, scikit-learn), SQL, R, Git
Personalization & Ranking	Candidate generation, ranking/reranking (rules + ML), feature design, offline evaluation (NDCG/MAP/recall@K as applicable), relevance-style analysis
Experimentation	A/B testing, hypothesis design, success metrics + guardrails, cohort analysis, statistical significance basics
Analytics & Modeling	Regression (linear/logistic/GLM), uncertainty estimation, Monte Carlo simulation, basic Bayesian modeling, time-series–style analysis
Data & Tools	Data pipelines, dashboards/BI-style reporting, Jupyter, GitHub workflows
Communication	Stakeholder communication, translating model results into product decisions, technical writing and presentations

EDUCATION

UNIVERSITY OF VIRGINIA **Charlottesville, VA**
PhD Candidate, *Astronomy* | Quantitative Modeling Expected May 2026
Master of Science, *Astronomy* | GPA: 4.00 Aug 2021 – May 2023

- Focus: high-dimensional data analysis, time-evolving systems, and probabilistic modeling using data from large observational facilities.
- Selected awards & leadership: NRAO Student Observing Support Award, Grote Reber Fellowship, Research & Professional Development Fellowship, and several teaching/academic awards (combined value $>$$75K); Founder & President, DIVERSITEA - departmental diversity organization; Co-founder & President, Pride & Ant-Prejudice - support group for international LGBTQ+ students

AGNES SCOTT COLLEGE **Decatur, GA**
BS, *Astrophysics*; *BA*, *Economics* | GPA: 3.98 Aug 2017 – May 2021

- Honors & awards: Summa Cum Laude, Phi Beta Kappa, Vedica Scholarship ($\approx$$240K), NASA Research Fellowship, NSF Research Fellowships (2018–2021).
- Leadership: President, South Asian Association; Founding Co-lead, Google Developer Students Club; Resident Assistant; Teaching Assistant.